MANOJ KALRA

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AUSTRALIA PERMANENT RESIDENT (PR) HOLDER

ELECTRONICS & COMMUNICATION ENGINEER

Smart Grid Testing SME with advanced experience in smart meter testing, and utility metering operations."

SUMMARY OF SKILLS

- Competent, diligent, and result-oriented professional with an **experience of 15 years** currently spearheading as **SME Testing with United Energy Melbourne**.
- Understanding of development process/ **SDLC (Agile/SAFe)** etc.
- **Experience in the field of Embedded system/ IoT based** utility products.
- Experience with **understanding the utility metering Firmware features and systems.**
- **Developing artifacts** i.e. Traceability Matrix, Test Plans, Defect Reports, Test Summary Report etc. for different testing activities for different meter and NIC firmware.
- Understanding on AMI DPI standards, ensuring consistent delivery and compliance with organizational frameworks.
- Applied advanced knowledge of AMI systems and Itron software to creatively solve technical challenges and deliver innovative solutions.
- **Understanding of firmware deployment in the field** and execution with AMI **C**ommunication **C**ontrol group **(ACCG)** for the firmware upgrade in the production meters.
- **Facilitating team meetings** with suppliers, vendors for defect triage, third-party solutions for test lab, regular or ad-hoc discussion etc.
- **System Engineering tasks** such as finalizing the User Stories, getting the review done and helping in releasing them.
- **Project management activities experience i.e. task listing, BOM prep, scheduling, execution, monitoring and status sharing, and closing etc.**
- Understand the **Epic, Features** and **User Stories** and write the **Test Cases** to validate them.
- Manage the Test Case Review process.
- Achieved key performance indicators (KPIs) and contributed to departmental objectives through strategic planning and execution of project initiatives.
- Creating **Test Suites** for different set of test cases and executing these test cases for new builds based on **Agile Methodology**.
- **Partially/fully Automation** of **Test Case Execution.**
- Executing **Sanity and Regression testing** for Application Builds**.**
- Proficient in **System Integration Testing.**
- Testing the product features with **Head End Systems, MDMS, OWOC/OWCE collection engines and integrate the automation testing.**
- Working experience on different protocols i.e. **UART, GPIB, ModBUS, CAN, TCP/IP, RS232.**
- Experience in handling different lab equipment and automating them e.g. **Programmable Power Supplies, Function Generator, Electronic Load, Oscilloscope, Thermal Chamber, Sniffer, JLink debugger, Renesas Debugger, MSP430 flasher, MSP430 debugger, Logic Analyzer/OTII** etc.
- Automate test solution for smart/IoT Utility Product i.e. Gas Meters, Electricity Meters and Water Meters.
- Experience in **verification** of **IoT Connectivity devices** i.e. smart utility meters with **cellular 4G CATM1** devices with **HL7800 Sierra Wireless Modem**
- Experience in activities such as preparing Bill of material **(BOM)**, Purchase Request **(PR)**, Purchase order **(PO)** for buying components for **ATE**.
- Understanding of **Hydraulic and Pneumatic circuits** for development of ATE.

- **Testing and Maintenance** activities for the production plant. Performing **Root Cause analysis**, **Preventive maintenance**, low downtime related activities.
- Successfully completed projects such as **Environmental Control System of Advanced Light Helicopter**, **Main Rotor Actuator and Tail Rotor Actuator Test Rig, UTF (Universal Test Fixture), ALT, ETA_Tool** and many more as part of work experience.
- Recipient **of "The Young Scientist" Award at National Level by Dr. A.P.J Abdul Kalam** also hold distinction of **scoring 94% (top 6%) at National Level in NITAT (2008)** by NIIT.

Technical Skills

Operating System:	Windows XP, Windows 7, Window 10
Electronics Workbench Tools:	**Certified LabVIEW Developer, NI LabVIEW, NI Test Stand (Maintenance Project), MCDWin, C#, DAQ MAX, PXI, PCI**
Instruments:	**Programmable Power Supplies, Function Generator, Oscilloscope, Thermal Chamber, Sniffer, Logic Analyzer/OTII**
Others:	**UIQ, Tortoise SVN, GIT, VSTS, Jenkins**

INDUSTRIAL EXPOSURE

United Energy (Melbourne, Australia) **June 2024 – Present**

Responsibilities:

- Responsible to perform end-to-end testing projects for the AMI meter firmware and NIC firmware to cover **DPI requirements** and additional functional requirements.
- **Providing support to Metering Operation Group (MOG)** for issue related to Local tools for meters i.e. EMPWIN+, Communication Tester, M-Cubed, Calegration etc. **Received Spotlight Award** in the categories "**Be The Best You Can Be**" and "**Succeed Together"** for providing support to MOG group.
- **Providing training to MOG group** for understanding of new introduced tool in the production.
- Fulfilling other teams' requirements in terms of meter configurations and providing them hardware resources for their testing.
- Execution of automating project for Lab Instruments for testing activities i.e. power supply, electronic load etc.
- Developing artifacts i.e. Traceability Matrix, Test Plans, Defect Reports, Test Summary Report etc. for different testing activities for different meter and NIC firmware.
- Raising defects for firmware issues and **coordinating with meters vendors** and **Itron** for different critical issues for the resolution.
- Guiding team members on **AMI DPI standards**, ensuring consistent delivery and compliance with organizational frameworks.
- Applied advanced knowledge of AMI systems and Itron software to creatively solve technical challenges and deliver innovative solutions.
- Working on firmware deployment plan with AMI **C**ommunication **C**ontrol group **(ACCG)** for the firmware upgrade in the production meters.

Tools: **HPALM, EMPWIN+, M-Cubed, Communication Tester, Calegration, LabVIEW, UtilityIQ (UIQ), SensorIQ (SIQ), MarketSystemTool/NetManager**
Electricity Meters: U-Series L&G Meters, Secure's PACT Meters, Secure's DLMS meters

Itron India Pvt Ltd **Nov 2015 – Oct 2022**
(~3years contract, ~4 years Permanent)

- To provide automated test solution for smart/IoT Utility Product i.e. Gas Meters, Electricity Meters and Water Meters.
- Experience in **verification** of **IoT Connectivity devices** i.e. smart utility meters with **cellular 4G CATM1** devices with **HL7800 Sierra Wireless Modem.**
- **Managing the SLNT testing project team** in terms of assigning and delivery of the work in global Cellular project.
- **Installation and configuration of Itron OpenWay Collection Engine environment** for different testing environments i.e. SLNT, Outcome etc.
- **Coordinated with third-party vendors to triage and resolve defects identified in external firmware components, including Sierra Modem firmware, ensuring timely issue resolution and system stability.**
- Understand the **Epic, Features** and **User Stories** and write the **Test Cases** to validate them.
- To manage the Test Case Review process.
- **Facilitated cross-functional collaboration with global development and testing teams to conduct reviews, manage defect triage, and drive timely resolution of issues.**
- Creating **Test Suits** for different set of test cases and executing these test cases for new builds based on **Agile Methodology**.
- **Automation** of **Test Case Execution** with automatic updating the test results in **VSTS.**
- Executing **Sanity and Regression testing** for Application Builds**.**
- Downloading/Erasing firmware from different microcontrollers and executing commands set on them to verify functionality of the firmware.
- Automatically triggering testing of firmware based on new firmware build and after complete execution of testing sharing html report with team.
- Worked with **LabVIEW OOPs** Concept.
- Developing **Python scripts** for parsing logic.
- Developing modules, need basis, on **C#**.
- Working knowledge with **32-bit microcontroller** for library testing i.e., **SPI, UART, Timer CSL libraries**
- Worked with different microcontroller i.e. **STM32L486RG, MSP430FR6922, MSP430G2403, RL78K-R5F104L30**
- Worked with different debuggers i.e. **JLink, RenesasE1, Flashpro430 flasher, MSP430 debugger**
- Used temperature chamber in automation to control environmental conditions for testing.
- Power Supply **Rigol DP832, Rigol DP711** used in automation (USB/ Serial Communication)
- Testing the product features with **Head End Systems, MDMS, OWOC/OWCE collection engines and integrate the automation testing.**

Freelancing (Melbourne, Australia) **June 2023 – June 2024**

Responsibilities:
- Automation solution development for lab Instruments.
- Perform testing for the Antennas for PassiveInterModulation and VSWR (Voltage Standing Wave Ratio)
- Developing automated solution for antenna testing using LabVIEW, Jira, Git, Confluence, C
- Driver development for different instruments i.e. PIM Instrument (AWT, Kaelus) and VSWR (S2 and TR from Copper Mountain)
- Integration of testing results to MES system
- Perform Factory Acceptance Testing.

ICT Group Netherlands (Eindhoven, Netherlands) **Nov 2022 – June 2023**

Project
Responsibilities:
- Development of Automated system for the Brewery process monitoring system.
- Using tools i.e. LabVIEW, Jira, Confluence, C#

- Instruments used- Flumbler with Embedded system having accelerometer mounted on the board to monitor the x,y, and z-axes to monitor the progress.

Medtronic PLC formerly Covidein Engineering Services P Ltd **January 2013 – Nov 2015**

- To understand the requirements and specifications of the product under development.
- Based on requirements develop the test cases to validate the product features.
- work on the Automated system development for the testing of the product.
- Selecting instruments, preparing BOM for the system.
- **Preparing project execution timeline** in terms of purchase, integration, execution, installation and commissioning etc. to PM.
- Planning and development from software (**LabVIEW/ C#**) and electronic/electrical integration to provide automated test solution.
- Preparing Validation document for the system.
- Preparing User manual for the developed system.
- Providing **end user training to use the automated solutions.**
- Completing **MSA (Measurement and System Analysis/ Gage R&R studies)** for the system.

Project

Company	**Covidien (Boulder-USA)**
Title	**Universal Test Fixture**
Role	Test Engineer
Tools	**LabVIEW**
Location	**USA**
Description	This Automated Test System is designed to simulate product environment condition and perform functionality of product in automation. Product was a Ligasure Vessel sealing device and aim of making the test system was to measure different functional parameter of the device.

Project

Company	**Covidien (Boulder-USA)**
Title	**Avengers**
Role	Test Engineer
Tools	**LabVIEW**
Location	**USA**
Description	This Automated Test System is design to perform insertion and extraction force of plug and receptacle of INVOS system. In this system 5 plug receptacle insertion extraction force was tested. Each plug receptacle could have maximum of 20 wires connection and total 100 wires resistance could also be measured on this fixture in automation. Force measurements and Resistance measurements were saved in CSV file.

Continental Automotive India Ltd **November 2011 – January 2013**

- Responsibility to manage for all Final Functional Tester based on **LabVIEW** and **MCDwin** software.
- To develop algorithm, updating and modification in running **LabVIEW and MCD codes** on Testers for reducing timing to increase production.
- Installation of new Testers for new Products.
- **Spare Parts Management**.
- **Calibration management** of Instruments.

Project

Company	**Hero Motocop**
Title	**Vision Inspection Station for KVNF**

Role	Executive Engineer
Tools	**MCDwin**
Description	This tester was developed to check all visual parameters of cluster of 2-wheeler bike. Like- pointer real-time check, display tests.

Company	**Hero Motocop**
Title	**Final Functional Tester for KVNF**
Role	Executive Engineer
Tools	**MCDwin**
Description	This tester was developed to check all parameters of cluster of 2-wheeler bike. Like- Real time speed, all display parameters.

MINISTRY OF DEFENCE **January 2011- November 2011**

Hardware Engineer

- Responsible for understanding the different **FEC** schemes and then implementing those schemes **on LabVIEW FPGA**.
- Responsible for testing those schemes for their error correcting capability.
- Troubleshooting when teammates using these FEC schemes.

Project

Company	**Indian Navy**
Title	**SDR (Software Define Radio)**
Role	Hardware Engineer
Tools	**LabVIEW 2009, LabVIEW FPGA, LabVIEW RTOS, Simulink**
Description	This project is based on developing different FEC scheme such as Convolution Encoder and Viterbi Decoder, RS Encoder Decoder etc.

NEOMETRIX ENGINEERING P LTD **August 2009 - December 2010**

Assistance Project Engineer

- Accountable for designing of full Electronic Circuits for test rigs (from field to data acquisition point).
- Independently supervising all activities including providing technical inputs for methodologies of installation and commissioning of Electrical Equipment & Systems with site management activities.
- Efficiently coordinating with other departments –Mechanical, Purchase, Logistics for smooth functioning of the projects.
- Understanding of sensors working such as: LVDT. Pressure Transmitter, Temperature Transmitter, RTD, Flowmeters, Load Cells as well as controller such as Servo Controller (MOOG).
- Responsible for making full software for Acceptance Test Procedure (ATP) for Unit under Test (UUT) on LabVIEW Platform. Also providing the customer user friendly GUIs so that full testing of that rig can be in full automation.
- Reviewing modification in electrical system as per process requirements & for reliability improvisations and offering customized solutions.
- Responsible for Data Acquisition through NI data Acquisition cards of sensors like LVDT. Pressure Transmitter, Temperature Transmitter, RTD, Flowmeters, Load Cells.

Projects

Company	**Hindustan Aeronautical Limited, Lucknow**
Title	**Environmental Control System of Advanced Light Helicopter Test Rig**
Role	Designing Electrical Circuits, Execution & Commissioning
Tools	**LabVIEW 8.2, PCI Card 6225, 6704 and Module DAQ- MAX**
Description	This Project involved test setup of ECS Units Aircraft's Air Conditioning for Advanced Light Helicopter in different environment and altitude.

Company	**Hindustan Aeronautical Limited, Lucknow**
Title	**Main Rotor Actuator and Tail Rotor Actuator Test Rig**
Role	Designing Electrical Circuits, Execution & Commissioning

Tools	**LabVIEW 8.2, PXI Card 6229, FRA 4461 and Module DAQ- MAX**
Description	This Project involved testing the MRA and TRA Unit of Helicopter while working on hydraulic controlled servo valve, servo amplifier, flow meter, Load cell, LVDT that was controlled by fully automated system. Various types of tests were performed such as Frequency Response Test, Linearity, Hysteresis, Leakage, Pilot Input Load Test and RAM Stroke.

EDUCATIONAL CREDENTIALS

Certification - **CLAD (Certified LabVIEW Associate Developer) Qualified**
CLD (Certified LabVIEW Developer) Qualified

M.Tech, **Embedded Systems (Work Integrated), India**
B.Tech, **Electronics and Communication Engineering, 2009, India**

XII 2004
Government Inter College, Hastinapur

X 2002
Government Inter College, Hastinapur

Projects undertaken
"Investigation of Friend or Foe" for Bharat Electronics Limited, a govt. of India (Ministry of Defence) Enterprise Bharat Nagar
Seminar Attended
Presented a seminar on "Wireless in local loop"
Presented a seminar on **"Kitchen Gardening and Nutrition"** at different villages **in class 11**

Extracurricular Accolades

- Received **Spotlight award** for the **Calegration** project delivery at **United Energy Melbourne**.
- Received '**Individual Excellence Award'** during my tenure with **Itron/Aricent**.
- Received **'Dream Team Award**' during my tenure with **Covidien**.
- Received '**Excellence Award'** award during my tenure with **Covidien.**
- Recipient of **"The Young Scientist"** Award at **National Level** by **Dr. A.P.J Abdul Kalam**
- Achieved **94 Percentile (top 6%)** at **National level in NITAT (2008)** by **NIIT**.
- Efficiently participated and secured **2nd position** in **Aptitude Test** organized at College Level.
- Distinction of having been awarded "**Certificate of Merit "**for academic excellence in college for **4 successive years**.
- Secured **2nd position** in C language test organized at College Level.
- Appointed as **Volunteer** in **Blood Donation Camp** for **Rotary Blood Camp** organized in our college.